Docker Volume:

Volumes are persistent data stores for containers, created and manageVolumes are persistent data stores for containers, created and managed by Docker.

When to use volumes

Volumes are the preferred mechanism for persisting data generated by and used by Docker containers. While bind mounts are dependent on the directory structure and OS of the host machine, volumes are completely managed by Docker. Volumes are a good choice for the following use cases:

- Volumes are easier to back up or migrate than bind mounts.
- You can manage volumes using Docker CLI commands or the Docker API.
- Volumes work on both Linux and Windows containers.
- Volumes can be more safely shared among multiple containers.
- New volumes can have their content pre-populated by a container or build.
- When your application requires high-performance I/O.

Mouting Local Storage With Docker Container:

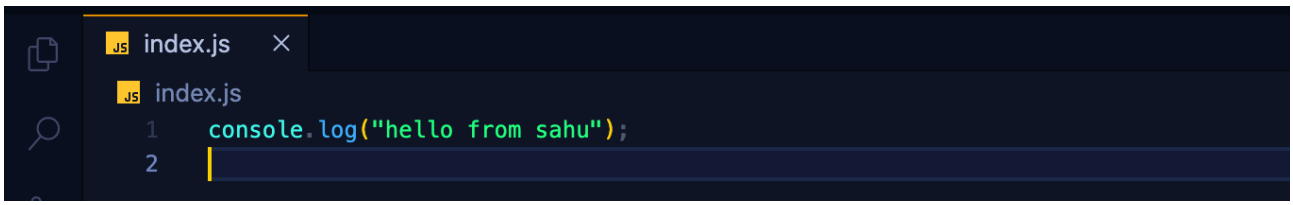
Step:

1.Create a empty folder in local storage in your laptop

2. Open the folder in code editor and make a file name index.js (file name can be any thing

In file write simple:

Console.log("hello from sahu");



3. Now run the command

Docker run -it node:latest

```
[ ✘ base sateeshsahu@Sateeshs-MacBook-Air ~/desktop docker run -it node:latest
Unable to find image 'node:latest' locally
latest: Pulling from library/node
b869a1ff74d6: Pull complete
0f3f14e70f1d: Pull complete
1cb11dd65b55: Pull complete
2aec02e53872: Pull complete
d00f6431566c: Download complete
26915215e68d: Download complete
Digest: sha256:3953ec6a2c10154a58ccf4ba48083ddfe3f8641d63f0d1d5cb8a4a78169123a7
Status: Downloaded newer image for node:latest
Welcome to Node.js v25.9.0.
Type ".help" for more information.
[>
```

Now we will attach local value with docker container:

Local location: */Users/sateeshsahu/Desktop/app*

Container Location: */home/app*

```
docker run -it -v /Users/sateeshsahu/Desktop/app:/home/app node bash
```

```
[ base sateeshsahu@Sateeshs-MacBook-Air ~ docker run -it -v /Users/sateeshsahu/Desktop/app:/home/app node bash
[root@52e4ea93cc26:/# ls
 bin  boot  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@52e4ea93cc26:/# cd home
[root@52e4ea93cc26:/home# ls
 app  node
[root@52e4ea93cc26:/home# cd app
[root@52e4ea93cc26:/home/app# ls
 index.js
[root@52e4ea93cc26:/home/app# node index.js
 hello from sahu
```

Mounting Docker volume with docker container:

First we will create a volume

Docker Command:

docker volume create ***volume-name***

We are making volume of name data

Docker volume create data

Now we will mount the docker volume with docker container

```
[ base sateeshsahu@Sateeshs-MacBook-Air ~ docker run -it -v data:/data node bash
[root@f997f1ae0e46:/# ls
 bin  boot  data  dev  etc  home  lib  media  mnt  opt  proc  root  run  sbin  srv  sys  tmp  usr  var
[root@f997f1ae0e46:/# cd data
[root@f997f1ae0e46:/data# ls
 sahu  satishsahu
[root@f997f1ae0e46:/data# mkdir devops
[root@f997f1ae0e46:/data# ls
 devops  sahu  satishsahu
 root@f997f1ae0e46:/data#
```